

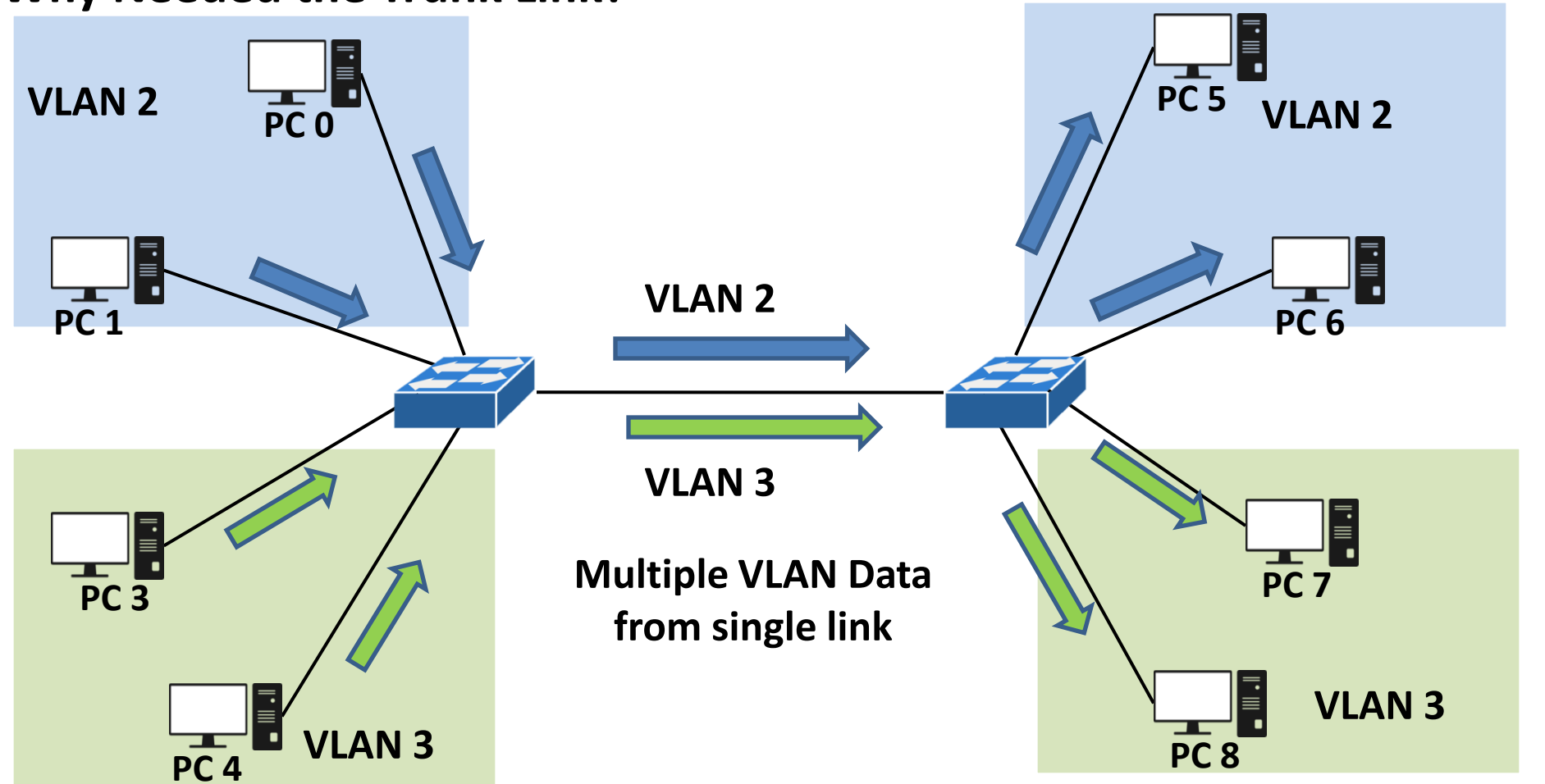
Trunking:

A **trunk link** is a type of network connection that carries traffic for multiple VLANs (Virtual Local Area Networks) between network devices such as switches and routers. Unlike access links, which carry traffic for only a single VLAN, trunk links use tagging mechanisms to ensure that frames from different VLANs are properly identified and routed to their correct destinations. Here's a detailed explanation of trunk links:

Key Characteristics of Trunk Links

- **Multi-VLAN Support**
- **Tagging Mechanism**
- **Inter-Switch Communication**

Why Needed the Trunk Link?



Trunk Commands

On Switch 0:

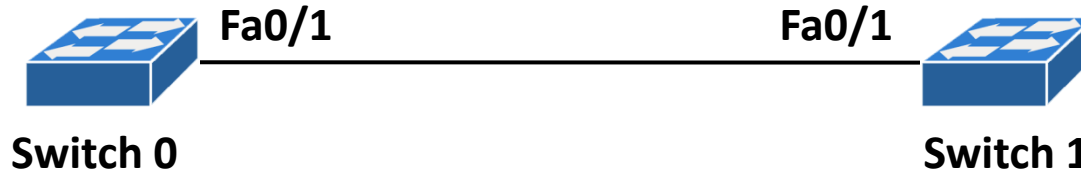
```
Switch(Config)#Int fa0/1
```

```
Switch(config-if)#Switchport Mode Trunk
```

On Switch 1:

```
Switch(Config)#Int fa0/1
```

```
Switch(config-if)#Switchport Mode Trunk
```



Verify Trunk:

Switch#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005

Port	Vlans allowed and active in management domain
Fa0/1	1

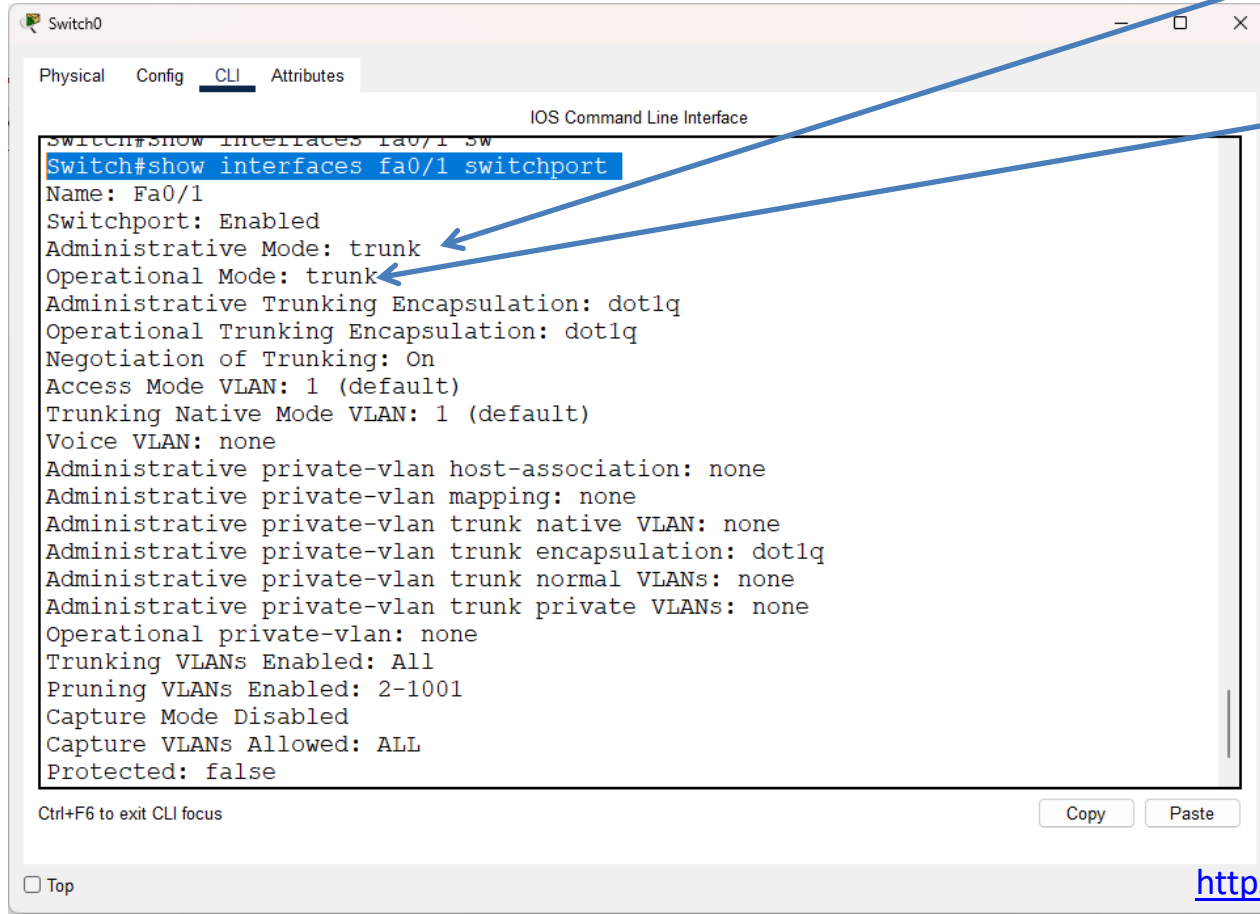
Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	none

Other Command to Verify interface:

Switch#show interfaces fa0/1 switchport

Interface Mode:
Administrative Mode

Link Mode:
Operational Mode



```
Switch0
Physical  Config  CLI  Attributes
IOS Command Line Interface
Switch#show interfaces fa0/1 sw
Switch#show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false

Ctrl+F6 to exit CLI focus
```